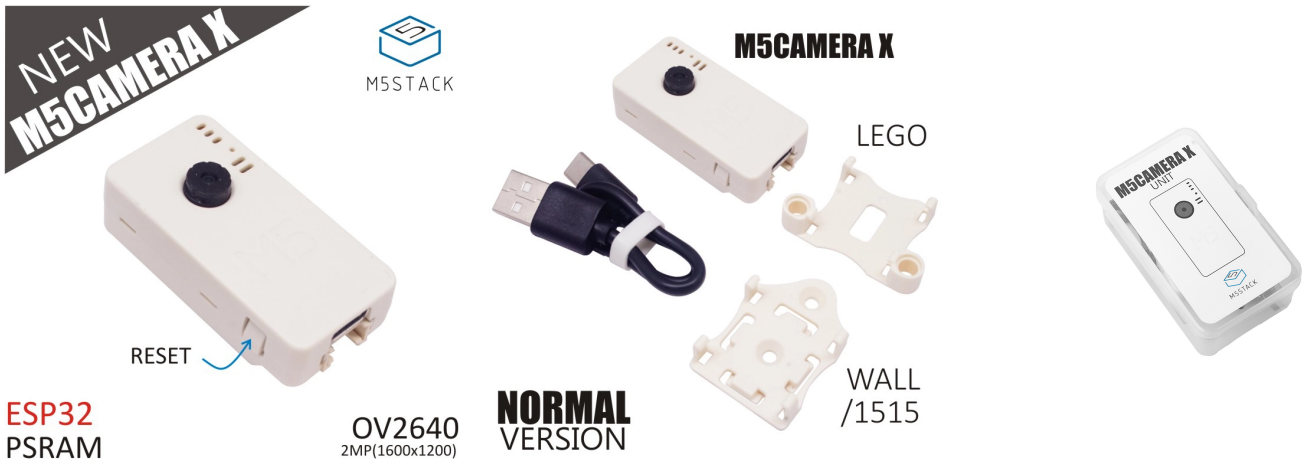
M5Camera-X

SKU:U038





Description

M5Camera-X is an image recognition development board that integrates the ESP32 (4M Flash + 520K RAM + 4M PSRAM) chip and a 2MP camera (OV2640). It supports Wi-Fi image transmission and USB port debugging.

The hardware comes preloaded with firmware, developed using ESP-IDF programming, running a Wi-Fi camera application. The default program outputs an image size of **600 x 800**, but you can optimize the program to output larger sizes.

How to use this program?

- Turn on the phone's Wi-Fi and scan to connect to an AP hotspot with a name starting with "m5stack-".
- Open the phone browser and visit 192.168.4.1 to enter the monitoring page and retrieve the live video.
- The video frame rate is about 5-6 frames per second.

Since the module can generate a Wi-Fi hotspot (AP), you can use a phone, PC, or other devices to wirelessly obtain camera images via Wi-Fi, or use the HY2.0-4P interface of the module to get camera images via a wired connection. It currently supports functions such as network camera, color recognition, and face recognition.

Features

- Designed based on ESP32
- Wi-Fi image transmission
- CP2104 USB TTL

Includes

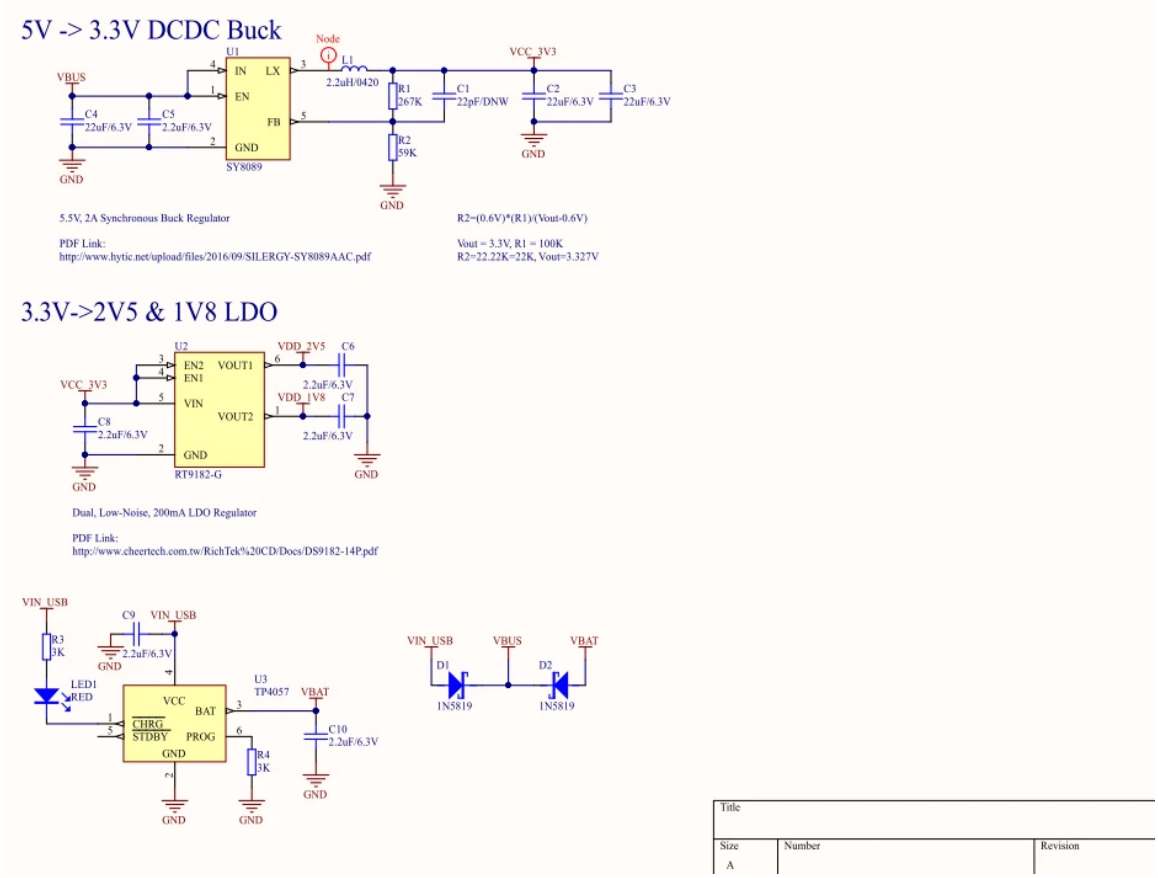
- 1 x M5Camera-X
- 1 x LEGO-compatible back clip
- 1 x Wall-1515
- 1 x USB Type-C Cable (20cm)

Specifications

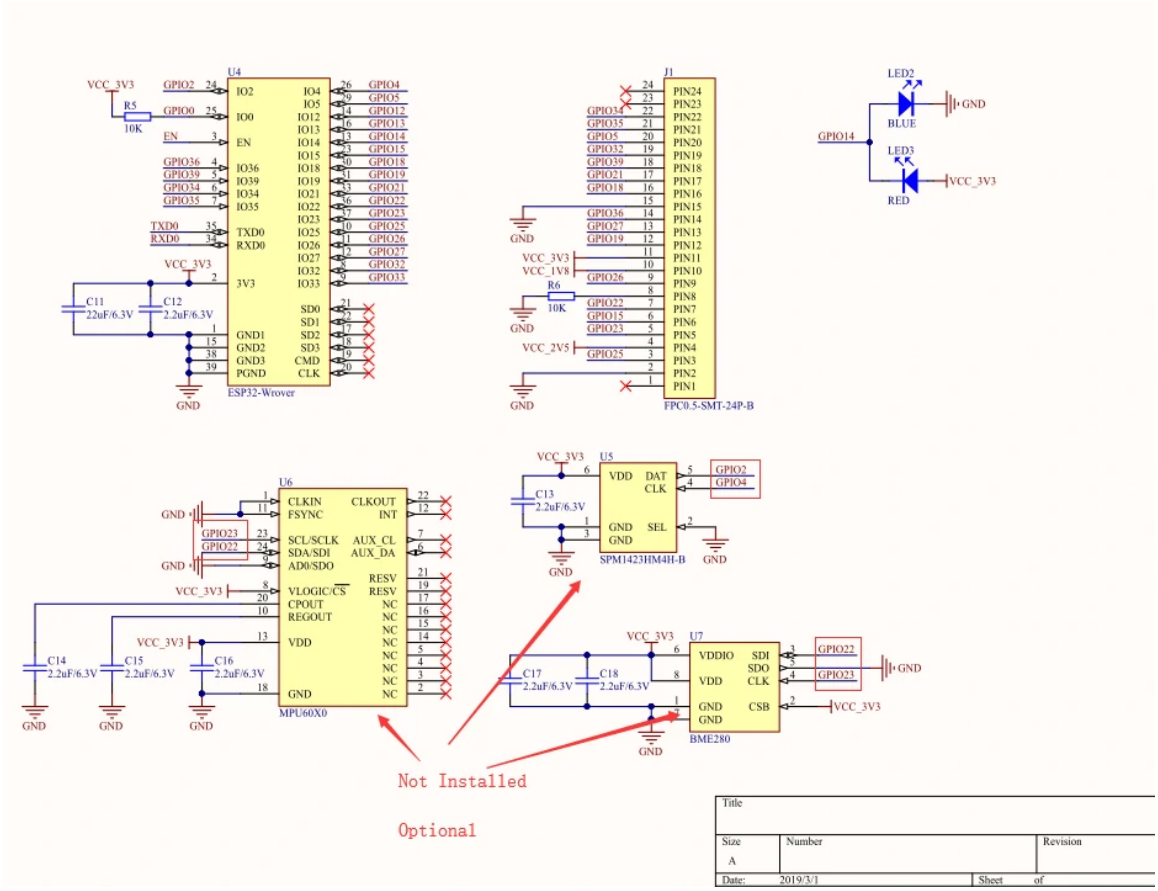
Specification	Parameter
RAM	4MB
Flash	4M
Image Sensor	OV2640
Max Resolution	2MP
Output Format	YUV (422/420) / YCbCr422, 8-bit compressed data, RGB565/555, 8-/10-bit Raw RGB data
Field of View	65°
Product Weight	14g
Gross Weight	38g
Product Size	24 x 48 x 13mm
Package Size	75 x 45 x 30mm

Schematics

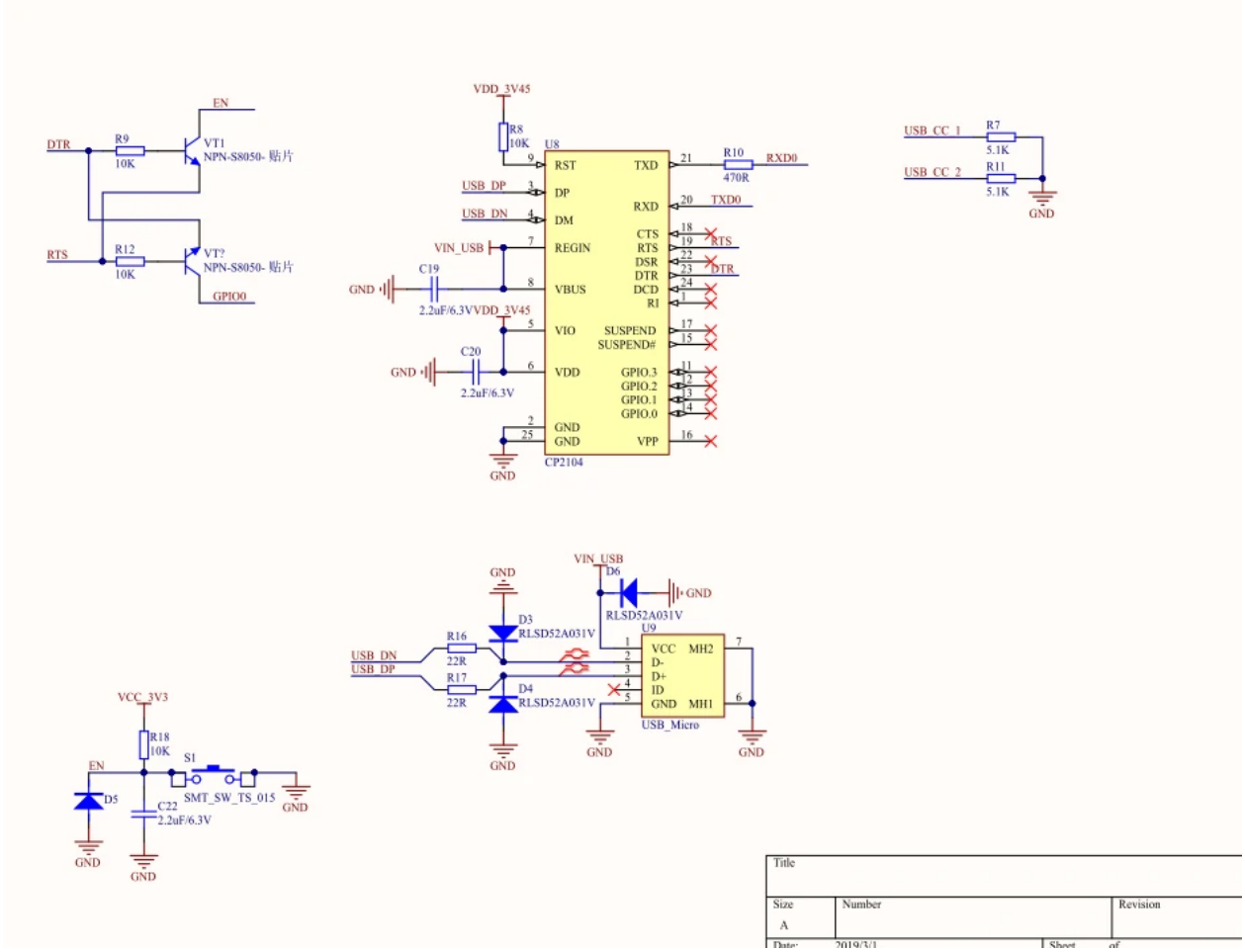
Power Circuit



Chip Peripheral Circuit



USB to Serial Circuit



PinMap

Camera Driver Chip OV2640 Interface

Interface	Camera Pin	M5Camera
SCCB Clock	SIOC	G23
SCCB Data	SIOD	G22
System Clock	XCLK	G27
Vertical Sync	VSYNC	G25
Horizontal Reference	HREF	G26
Pixel Clock	PCLK	G21
Pixel Data Bit 0	D2	G32
Pixel Data Bit 1	D3	G35
Pixel Data Bit 2	D4	G34
Pixel Data Bit 3	D5	G5
Pixel Data Bit 4	D6	G39
Pixel Data Bit 5	D7	G18
Pixel Data Bit 6	D8	G36
Pixel Data Bit 7	D9	G19
Camera Reset	RESET	G15
Camera Power Down	PWDN	**See Note1**
Power Supply 3.3V	3V3	3V3
Ground	GND	GND

Note1

The **PIN8 (PDWN)** pin of the OV2640 chip is the enable pin. On this board, it is connected to ground through a 12KΩ pull-down resistor to enable and enter working mode. When the PIN8 (PDWN) pin is pulled high, it will enter **Camera Power Down** mode.

HY2.0-4P Interface

HY2.0-4P	M5Camera
SCL	G13
SDA	G4
5V	5V
GND	GND

LED Interface

LED	M5Camera
LED_Pin	G14

The following are PCB reserved interfaces, without actual soldered components.

BME280 Interface

BME280	M5Camera
SCL	G23
SDA	G22

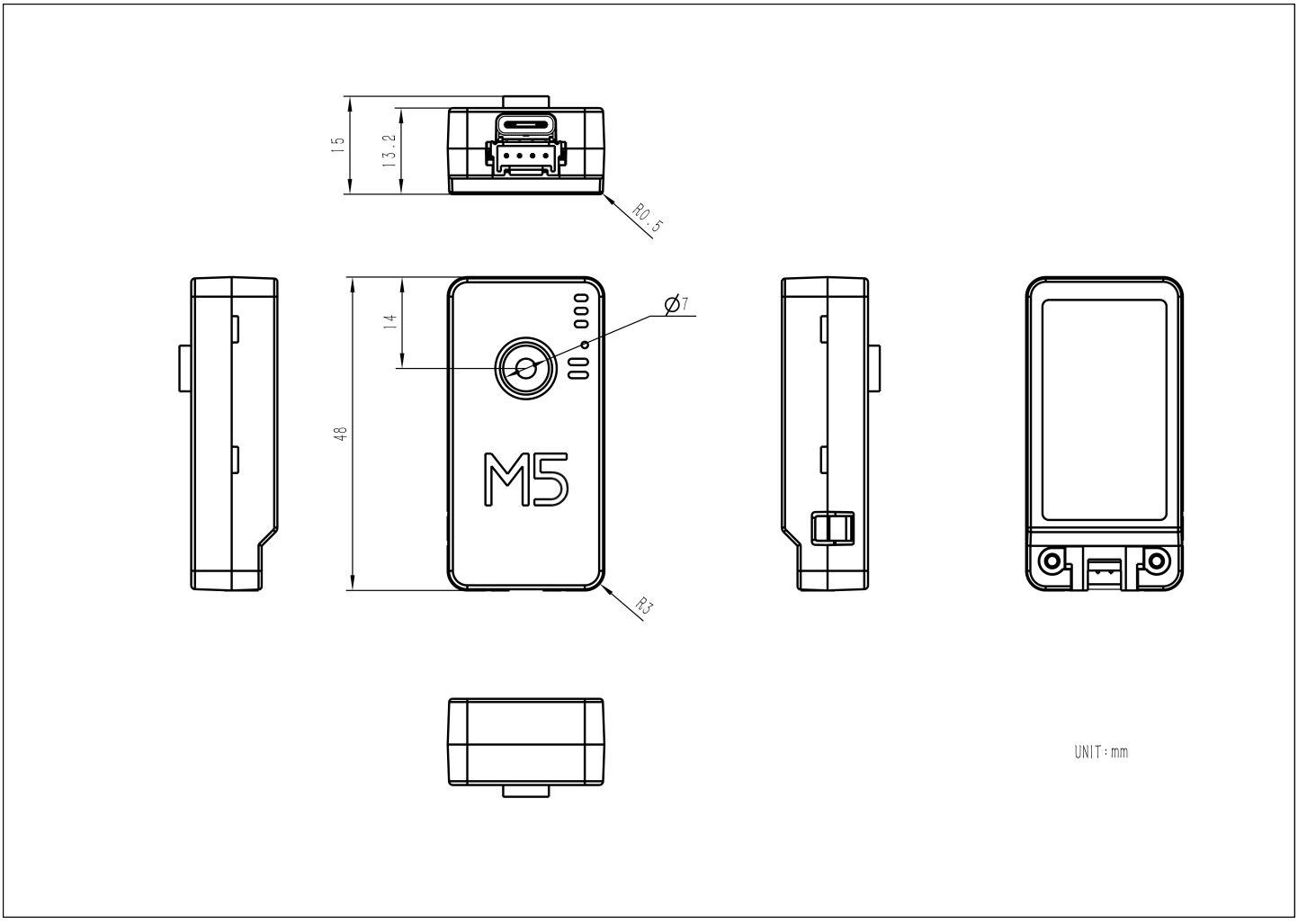
MPU6050 Interface

MPU6050	M5Camera
SCL	G23
SDA	G22

MIC (SPM1423) Interface

MIC (SPM1423)	M5Camera
CLK	G4
DATA	G2

Model Size



Structure

- M5Camera-X Structure Files

Datasheets

- ESP32
- OV2640

Softwares

Arduino

- M5Camera-Arduino

ESP-IDF

- M5Camera
- M5CameraX Firmware
- Face Recognition
- Serial Communication - M5CameraX
- Serial Communication - M5Core
- QR Code Recognition
- MPU6050

Easyloader

Easyloader	Download	Note
M5Camera-X AP Firmware Easyloader	download	/

Product Comparison

- View M5 Camera Series / Product Difference